Gagana M D

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AI safety researcher working on monitoring and evaluating AI agents. I build controlled experiments and the evaluation code behind them (frozen protocols, tests, preserved run artifacts), and I report what the evidence does and doesn't support.

Research Experience

PRISM AI Safety Research Fellowship May 2026 – Present
Research Fellow and Research Support – Remote

- **LLM-generated world models** (mentor: Jobst Heitzig). Main code contributor (78 of ∼116 commits) to a 44-scenario benchmark of LLM-written executable POSGs across five LLMs. 134/135 games passed mechanical checks, yet mean structural F1 was 0.37. Paper under review at a NeurIPS 2026 workshop.
- Research support for two teams, plus experiment tracking and reviews. **Causal-reasoning interpretability** (mentor: Manjari Narayan): built the pipeline (synthetic causal vignettes, prompt-stability analysis, layerwise probes on Qwen2.5). **Synchronous monitoring for AI control** (mentor: Preeti Ravindra): extending experiments against established baselines.

BaseThesis Labs Apr 2026 – Present
AI Safety Researcher, Evaluations and Benchmarking – Remote

- Built an offline-first benchmark harness for long-horizon finance and diligence agent tasks: rubric grading with safety dealbreakers, a leakage guard, trace capture and state-diff auditing; 50 tests, CI. Code.
- Measured when agent orchestration pays off (coding, reconciliation, a 717-item diligence corpus). It cost 3.5× latency and helped only where coverage and traceability mattered. Code.
- Leading **Secret Loyalties**: can hidden objectives in language models be recovered from observable evidence? (in progress)

Sentient Futures Sep 2026 – Present
Research Fellow (mentor: Julia Bossmann) – Remote

- **Gap Attractor:** behavioural evaluations of whether LLMs show stable, self-directed curiosity across prompting conditions and controls.
- **Post-Control Initiative:** lead interdisciplinary roundtables (AI safety, governance, game theory) on cooperation when human control can't be assumed.

FlowResearch Apr 2026 – Present
Research Fellow: behavioural evaluation of RL and world-model-based agents – Remote

Indian Institute of Science, DREAM Lab Jan 2025 – Aug 2025
Research Intern – Bengaluru

- RL policies for splitting inference across edge devices (ExpansionNet v2, YOLOv5); distributed optimisation that improved system utility by 31%.

Neuromatch Oct 2024 – Mar 2025
Impact Scholar: co-authored a study of the Matching Law in transformer-based RL agents – Remote

Selected Project

Detection Is Not Containment – Apart Research AI Incident Response Sprint Sep 2026

- First author; built the monitoring experiment. Does a monitor flag a prohibited agent action early enough for revoking authority to prevent it? Hash-frozen oracle and prompt, benign-only threshold calibration, preregistered phases, deterministic tests.
- Detection preceded the last safe point by one step; meaning-preserving renames kept only 8/12 decisions; a replayed false alarm failed a benign task. Case study, code.

Publications and Reports

Gagana M D, Ajay Kumar K, Vijayashekhar S S, Jignesh S. Bhatt. *Hyperspectral BiorNet: A Complete Wavelet Framework for Unmixing.* IEEE InGARSS 2025, pp. 15–19. DOI

Gagana M D, Kaustubh Gupta, María Inés Berdiñas. *Detection Is Not Containment: Timely Monitoring Determines Whether AI Agent Failures Can Be Prevented.* Apart Research, 2026. Report

Y. Karakoc, E. Sharifi, **M. D. Gagana**, Z. Sarayloo. *Investigating the Matching Law in Transformer-Based Reinforcement Learning Agents.* Zenodo, 2025. DOI

Paper on the structural fidelity of LLM-generated world models. Under review, NeurIPS 2026 workshop.

Education

B.E., Artificial Intelligence and Machine Learning – Acharya Institute of Technology, CGPA 8.38/10 2021 – 2025
Also: Neuromatch Academy, Computational Neuroscience (2025) and Deep Learning (2024).

Honours, Skills and Community

Honours: 2nd place, AMD AI Sprint Hackathon (AMD, Meta, Hugging Face), adversarial Q&A agents; 1st place, Federated Learning Client Selection Challenge, IISc; 1st place, Srushti 2025 state-level project exhibition.
Tools: Python, PyTorch, Transformers, Stable-Baselines3, Ollama, pytest, GitHub Actions, Docker, FastAPI, Git, Linux.
Community: Women in Big Data India, Shakti Mentorship Program team and workshop facilitator (2026–); WiDS Ambassador (2026–); Women Who Code Data, helped run 11+ workshops reaching 350+ people (2022–24).